SIVARAJ V

+91-9384969886 | sivarajvofficial@gmail.com | linkedin.com/in/sivarajvofficial | github.com/sivarajv04 | Portfolio

PROFESSIONAL SUMMARY

Final year B.E. Computer Science Engineering student and aspiring AI/ML Engineer with production-grade project experience in generative AI, computer vision, and retrieval-augmented generation. Trained StyleGAN2 models on 100,000+ images, built agentic LangGraph pipelines, and deployed end-to-end ML systems serving live users. Completed internships at Zoho Corporation and Infosys Springboard with measurable production impact. Available for full-time roles from mid-2026.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript
ML & Deep Learning: PyTorch, Scikit-learn, XGBoost, TensorFlow, CNN, Transfer Learning, Hyperparameter Tuning
Generative AI / LLM: RAG Pipelines, LLM Integration, GAN, Prompt Engineering, FAISS, Embedding Models
Computer Vision: OpenCV, Image Classification, ROI Extraction, Defect Detection, Preprocessing Pipelines
Backend & APIs: FastAPI, REST APIs, Async Processing, Microservices, ReactJS
MLOps & Tools: Weights & Biases, Experiment Tracking, Git, Docker (basics), CUDA, GPU Training (T4)
Statistical foundations: probability, hypothesis testing, model evaluation metrics
Databases: PostgreSQL, FAISS, SQL Optimization, Schema Design
Deployment: Vercel, HuggingFace Spaces, Distributed Multi-Service Architecture

EXPERIENCE

Zoho Corporation Pvt. Ltd. — *July 2025 – August 2025*
Software Developer Intern — *Tamil Nadu, India*

- Designed and deployed CRM automation services processing **500+ records daily** via Deluge scripting and API integrations, replacing a fully manual data handling workflow.
- Restructured validation and automation logic, **cutting manual operational effort by 70%** and removing repeated human review from the daily processing cycle.
- Implemented production error handling, retry mechanisms, and structured logging, reducing unhandled failures in live environment.

Infosys Springboard — *December 2025 – Present*
AI Application Intern — *Remote*

- Built end-to-end computer vision pipeline using PyTorch and OpenCV for automated PCB defect detection, replacing manual visual inspection with a fully automated inference system.
- Designed preprocessing pipeline covering XML annotation parsing, ROI extraction, and multi-class classification, **improving defect detection clarity by 40%**.
- Evaluated model performance using precision, recall, F1-score, and confusion matrix across repeated test cycles for reproducible inference behavior.

PROJECTS

Personix AI – Synthetic Face Dataset Generation Platform — **Live Project: personix-ai.vercel.app**
PyTorch | StyleGAN2-ADA | FastAPI | PostgreSQL | ReactJS | Weights & Biases | Vercel

- Built a full-stack generative AI platform for ordering configurable synthetic face datasets by age, gender, and volume, replacing manual dataset curation entirely.
- Trained StyleGAN2-ADA on **100,000+ curated images** using CUDA T4 GPU, monitoring loss convergence and augmentation via Weights & Biases to stabilize GAN training.
- Designed threshold-based inventory automation that self-triggers generation jobs when stock drops below threshold, **eliminating all manual regeneration steps**.
- Deployed distributed multi-service architecture (frontend, backend API, model inference, database) enabling independent scaling with zero-downtime updates.

PCB Defect Inspector – Automated Optical Inspection System — **Live Project: HuggingFace Spaces**
PyTorch | OpenCV | Multi-Model Classification Pipeline | HuggingFace Spaces

- Developed and deployed an automated optical inspection app detecting PCB manufacturing defects using a multi-model deep learning pipeline, replacing manual board review.
- Benchmarked 3 classification models achieving **99% validation accuracy**; selected optimal model via precision-recall-F1 evaluation across all defect categories.
- Refined ROI extraction and multi-stage validation, **reducing false positives by 60%** vs. single-model baseline, improving reliability for production-quality inspection.

RAG Document Intelligence System – Retrieval-Augmented Generation Backend — *In Progress*
FastAPI | PostgreSQL | FAISS | Vector Embeddings | LLM Integration

- Building an end-to-end RAG pipeline enabling natural language document queries, replacing manual search with embedding-based semantic retrieval over structured document collections.
- Implemented FAISS vector indexing and similarity search, **reducing document lookup time by 90%** in internal testing vs. keyword-based manual search "state-of-the-art" .
- Built modular FastAPI backend with PostgreSQL schema for traceable query logging and extensibility for future agentic workflow integration.

EDUCATION

SCAD College of Engineering and Technology — *2022 – 2026*
B.E. Computer Science and Engineering — CGPA: 7.9 / 10